Non-Redundant and Symmetry-Adapted Coordinates for Black-Box Molecular Structure Refinement

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Abstract

Semiexperimental equilibrium structures combine high-resolution rotational spectroscopy with computed vibrational and electronic corrections. The least-squares refinement depends on the coordinate model used to connect observables, constraints, and structural parameters. Z-matrices are compact for simple acyclic molecules; in rings, fused systems, and bridged frameworks they introduce reference-tree choices, derived ring quantities, and manually selected symmetry relationships.

We present `SEfit`, a black-box structure-refinement framework in which the input structure is Cartesian and the active coordinate model is generated from molecular topology and symmetry. Two representations are available. The first is a non-redundant symmetry-adapted generalized-internal-coordinate basis built from homogeneous primitive blocks of stretches, bends, torsions, out-of-plane motions, and ring coordinates. The second is a symmetry-adapted Cartesian displacement basis obtained after exact removal of translations and rotations. In both representations all atoms

enter through the same Cartesian topology, and only the totally symmetric active subspace is refined.

The chemical interface remains internal-coordinate based. Constraints, predicates, and parameter classes are specified as primitive relations because bond lengths, valence angles, dihedrals, and local hydrogen frames are the quantities that carry chemical meaning. These relations are projected onto either active basis through the Wilson matrix or the corresponding primitive-coordinate derivative projector, preserving chemically defined constraints and rigorous uncertainty propagation.

`SEfit` integrates moment-of-inertia fitting, vibrational and electronic corrections, predicate observations, parameter classes, reduced-dimensionality hydrogen models, statistical diagnostics, and human-readable reports within a single Cartesian/topology-based refinement workflow.

Glycolaldehyde is used as a standard acyclic reference. Glycine II tests functional constraints inherited from high-level structural information, whereas cyclopentadiene, nitrobenzene, azulene, and norcamphor probe cyclic, planar aromatic, polycyclic, and bridged cases. Across these benchmarks the generalized-internal-coordinate and symmetry-Cartesian refinements give the same numerical ranks, effective dimensionalities, and rotational residuals, while final structural parameters and uncertainties remain chemically interpretable.

1 Introduction

Accurate equilibrium molecular structures are central to rotational spectroscopy, thermochemistry, force-field development, and the validation of electronic-structure methods.[1,2] More generally, structure determination from experimental observables is a coordinate-dependent inverse problem. The data constrain a Cartesian geometry, but the fitted variables determine conditioning, constraints, correlations, and propagated uncertainties. Rotational spectroscopy is a stringent example because rotational constants probe the molecular mass distribution with high precision, whereas semiexperimental (r_e^{SE}) structures require vibrational and sometimes electronic corrections to convert ground-state constants into equilibrium information.[3–6]

Modern semiexperimental and rotational-structure fitting programs and workflows, including the `MSR` molecular-structure-refinement program and the `STRFIT` structure-fitting program distributed with `PROSPE`,[7–9] have established robust statistical principles: fit moments of inertia or otherwise well-conditioned observables, include theoretical predicates when experimental information is incomplete, and propagate the covariance matrix to derived structural parameters.[10] The remaining bottleneck is the coordinate model rather than the least-squares formalism. This issue is easily hidden in small acyclic molecules, where a single Z-matrix reference tree can represent the important distances and angles without serious ambiguity.

Recent automated workflows have substantially extended the practical reach of semiexperimental refinement by coupling computed starting geometries, efficient second-order vibrational perturbation theory (VPT2)-based vibrational corrections, and reduced-dimensionality refinements in a largely black-box pipeline.[11] `SEfit` uses such computed geometries and correction data as upstream inputs and replaces the remaining user-defined Z-matrix layer with a topology-driven coordinate engine.

The same construction is not unique for rings, fused systems, and bridged frameworks. A Z-matrix is a tree representation of a molecular graph, whereas many molecular skeletons are not trees. Closing a ring therefore requires that at least one topological distance, angle, or dihedral be derived from the remaining variables rather than treated on the same footing. Different atom orders or reference paths then define different statistical models: one bond may be an adjustable parameter in one Z-matrix and a derived quantity in another, even though both describe the same Cartesian structure. Symmetry relationships and hydrogen constraints can compensate for this non-uniqueness, but only if they are introduced manually and consistently.

This work replaces the user-defined coordinate model by an automatic topology-driven layer. Starting from Cartesian coordinates, the coordinate engine builds primitive internal coordinates, removes redundancies, assigns symmetry, and supplies `SEfit` with the totally symmetric active space. Two coordinate choices are available. The default is a symmetry-adapted GIC basis constructed within homogeneous coordinate families, so stretches, bends, torsions, out-of-plane motions, and ring coordinates are not mixed indiscriminately. The alternative is a symmetry-adapted Cartesian basis obtained by projecting out translations and rotations and then applying the same point-group selection. Both are black-box topology-derived non-redundant coordinate models: every atom enters through the same Cartesian topology, no traversal path is selected, and no local structural parameter is privileged by construction.

The chemically meaningful interface remains internal-coordinate based. Constraints, predicates, and parameter classes are defined as primitive internal-coordinate relations and then projected onto whichever active basis is used. This is essential for the Cartesian option: a fixed C–H bond, an X–Y–H angle, or a local hydrogen frame is simple in primitive internal coordinates, whereas an equivalent Cartesian constraint is origin- and orientation-dependent, nonlocal, and generally not chemically transparent. The projection therefore uses the Wilson matrix or the corresponding primitive derivative operator[12] rather than asking the user to define Cartesian component constraints.

The rotational semiexperimental problem is used here as a demanding test of a broader question: which coordinates provide the most robust route from experimental observables to molecular structures? Glycolaldehyde represents a standard case where conventional Z-matrix methods are already adequate. Glycine II, cyclopentadiene, nitrobenzene, azulene, and norcamphor are deliberately less forgiving benchmarks because they require incomplete isotopic information, functional structural constraints, ring closures, planar-component selection, fused or bridged topology, and nontrivial constraints. The results show that automatic GICs and symmetry-adapted Cartesians lead to the same effective spaces and residuals, while final bond lengths, angles, dihedrals, and their uncertainties are reported uniformly from the optimized Cartesian geometry.

2 Theory and Method

2.1 Overview of the SEfit Workflow

The present framework has three layers. The first is the chemical interface: Cartesian geometry, topology, primitive internal coordinates, and any constraints, predicates, or parameter classes. The second is the active coordinate basis, chosen either as symmetry-adapted GICs or as symmetry-adapted Cartesian displacements. The third is the `SEfit` least-squares backend, which is independent of how the active basis was constructed.

This separation is important. The user specifies chemically meaningful relations in primitive internal coordinates, while the numerical solver sees a non-redundant totally symmetric active space. The final output is again chemical: optimized Cartesian coordinates, primitive bond lengths, angles, dihedrals, and propagated uncertainties.

The starting Cartesian geometry can be supplied by experiment, by any quantum chemical model, or by an automated composite-scheme generator. These starting geometries can also provide reference values for hard constraints, predicates, and local hydrogen frames when the isotopic data are insufficient to determine all hydrogen coordinates independently. The coordinate engine keeps the covalent topology used for primitive coordinates, ring detection, redundancy removal, and symmetry adaptation separate from any auxiliary reference information used to define constraints or predicates.

The overall architecture is summarized in Figure 1.

This separation between chemical specification, active coordinate basis, and least-squares refinement allows the same infrastructure to support both semiexperimental structure determination and other coordinate-dependent modules.

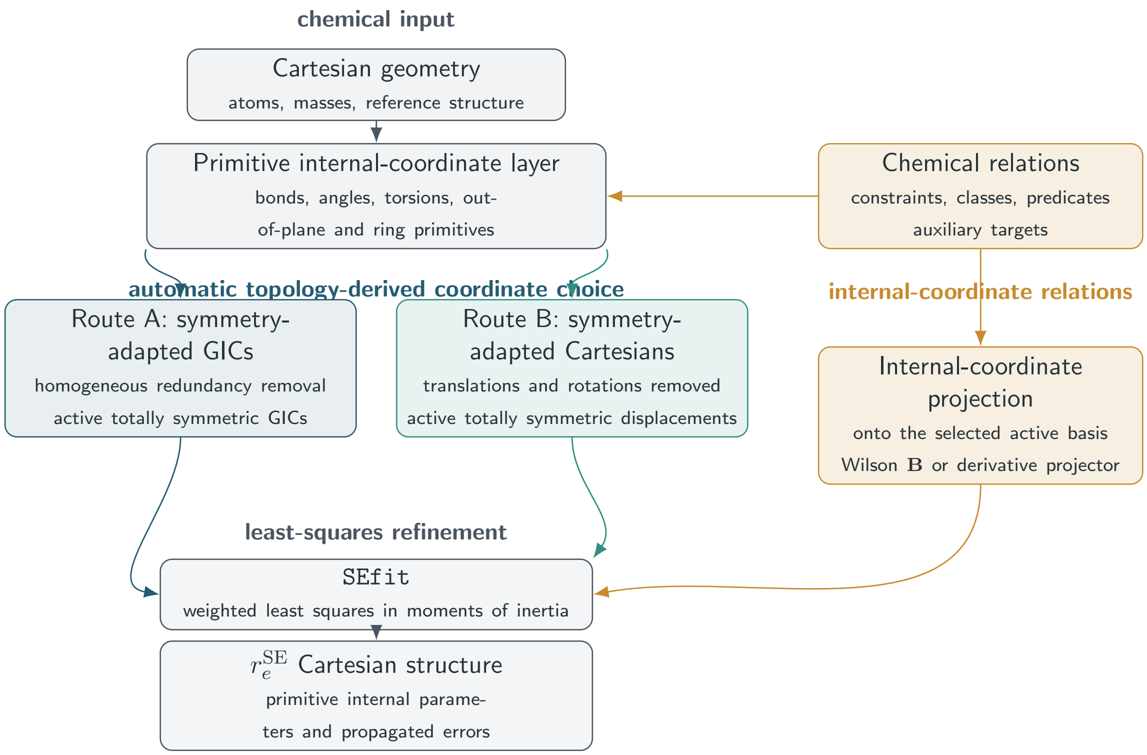

Figure 1: Architecture of the automatic `SEfit` workflow. Cartesian geometry defines topology and primitive internal coordinates once. The active least-squares space is then chosen as either Route A, a non-redundant symmetry-adapted GIC basis, or Route B, a symmetry-adapted Cartesian basis with translations and rotations removed. Constraints, parameter classes, and predicates remain internal-coordinate objects and are projected onto either active basis before refinement. The final Cartesian structure and the propagated errors of primitive internal parameters are generated by the same reporting layer. Auxiliary model information is handled in the chemical-relation layer and is not inserted into the covalent graph used to generate GICs.

2.2 Primitive Coordinate Generation

The coordinate construction starts from a Cartesian molecular geometry. The coordinate engine first perceives the covalent graph, including bonded atom pairs and ring connectivity. This graph is the only topology used for primitive-coordinate generation, ring detection, redundancy removal, and symmetry adaptation.

The graph is then expanded into a deliberately redundant primitive set: stretches, valence angles, torsions, out-of-plane deformations, linear bends, and ring-specific primitives. Redundancy at this stage is useful because it preserves locality and avoids forcing a premature choice of which primitive coordinates should become fitted variables.

Auxiliary reference information is kept outside the covalent graph. It may define constraints, predicates, or starting-geometry corrections, but it is not inserted into primitive GIC generation or ring detection.

For cyclic systems, ring coordinates are generated from endocyclic torsions and from analogous combinations of endocyclic valence angles. Unlike Cremer–Pople variables[13] or manually defined ring coordinates, these primitives remain within the generalized-internal-coordinate formalism and are available for monocyclic, fused-ring, bridged, and polycyclic systems.

This primitive layer is also the user-facing chemical interface. Constraints, predicates, and parameter classes are expressed here and remain independent of the non-redundant active representation generated later.

2.3 Coordinates for Parameter Refinement

Coordinate requirements differ between geometry optimization and parameter refinement. Redundant coordinates are often useful in optimization because the optimizer can work in an overcomplete space and project out near-null directions. In a least-squares refinement, however, the adjustable variables are the statistical parameters of the model. Their normal matrix must have a well-defined rank, their covariance matrix must be meaningful, and that covariance must be propagated to reported bond lengths, angles, dihedrals, and other derived quantities. The active space must therefore be non-redundant, symmetry consistent, and compatible with chemical constraints.

The non-uniqueness of Z-matrices is central in this context. A Z-matrix is not only a coordinate transformation; it also selects a rooted traversal of the molecular graph. For an acyclic graph this choice can often be made without major chemical consequences. For a ring or a bridged skeleton, however, the graph contains cycles and no tree can place all chemically equivalent edges on the same footing. The ring-closing edge is necessarily represented indirectly, and a different atom order changes which coordinate is primary and which is derived. The fitted covariance matrix is therefore attached to a particular coordinate model, not simply to the molecule.

We call a coordinate construction democratic when it satisfies three conditions. First, it starts from the complete Cartesian topology rather than from a user-selected reference path. Second, all topologically implied primitive coordinates of a given type are generated before any reduction is performed. Third, symmetry and rank reduction are applied algorithmically, so equivalent local fragments are treated identically unless the experimental model explicitly introduces a symmetry-allowed constraint or parameter class. This definition does not require the final working coordinates to be local chemical coordinates; it requires the construction of the active space not to privilege one atom traversal, ring edge, or fragment over another.

The first active space satisfying these requirements is a symmetry-adapted GIC basis. The coordinate engine builds standard primitive internal coordinates, including bond lengths, valence angles, torsions, linear bends, out-of-plane deformations, and ring coordinates derived from endocyclic torsional and valence-angle primitives. Redundancy removal and symmetry adaptation are carried out in homogeneous blocks, so stretches are combined only with stretches, bends only with bends, and torsional or ring coordinates only with coordinates of the same physical type. This preserves locality while eliminating linear dependencies.[14–17]

The second active space is a symmetry-adapted Cartesian displacement basis. Translations and rotations are projected out, the remaining vibrational Cartesian space is symmetry adapted, and only the totally symmetric block is used. This basis has no direct local chemical meaning, but it is complete, non-redundant, and independent of a Z-matrix. Since final structural parameters are always evaluated from the optimized Cartesian geometry and their uncertainties are propagated to primitive internal coordinates, local chemical interpretation is recovered at the reporting stage.

Symmetry plays two separate roles. It reduces the active space to the totally symmetric representation, ensuring that the refinement cannot lower the point group of the parent structure. It also completes user-specified primitive constraints over symmetry-equivalent orbits, so a constraint on one member of an equivalent set cannot break the symmetry of the fit. Non-totally-symmetric coordinates remain part of the complete coordinate report and are useful for diagnostics, but they are inactive in an equilibrium refinement.

2.4 Non-Redundant Symmetry-Adapted GICs

The redundant primitive-coordinate set generated from molecular topology provides a chemically intuitive description of molecular structure but contains linear dependencies arising from graph connectivity, ring closure conditions, and molecular symmetry. These dependencies are removed before refinement so that the normal equations, covariance matrix, and propagated parameter uncertainties are defined in a well-conditioned independent coordinate space.

Let $\mathbf{x} \in \mathbb{R}^{3N}$ be the Cartesian coordinate vector and let

$$\mathbf{s}(\mathbf{x}) = (s_1(\mathbf{x}), s_2(\mathbf{x}), \ldots, s_m(\mathbf{x}))^T \tag{1}$$

collect the primitive internal coordinates. At a reference geometry $\mathbf{x}_0$, primitive displacements and Cartesian displacements are related to first order by the Wilson matrix[12]

$$\Delta \mathbf{s} = \mathbf{B}_s \Delta \mathbf{x}, \qquad \mathbf{B}_s = \left.\frac{\partial \mathbf{s}}{\partial \mathbf{x}}\right|_{\mathbf{x}_0}. \tag{2}$$

The primitive vector is partitioned into homogeneous families,

$$\mathbf{s} = (\mathbf{s}_{\text{str}}, \mathbf{s}_{\text{bend}}, \mathbf{s}_{\text{tor}}, \mathbf{s}_{\text{oop}}, \mathbf{s}_{\text{ring}}, \ldots), \tag{3}$$

The reduction to non-redundant GICs is performed before symmetry adaptation and is based on the rank of the primitive Wilson matrix. The procedure is not a selection of a chemically convenient subset of primitives. It is a linear rank-revealing transformation of the full primitive set. The blocks are processed in a fixed topological order, and all rank decisions use the same singular-value thresholds; thus the unsymmetrized basis is determined by the Cartesian geometry and molecular graph, not by the subsequent fit. For each coordinate family α, the corresponding block of the Wilson matrix is

$$\mathbf{B}_{s_\alpha} = \frac{\partial \mathbf{s}_\alpha}{\partial \mathbf{x}}. \tag{4}$$

A singular-value decomposition,[18]

$$\mathbf{B}_{s_\alpha} = \mathbf{U}_\alpha \mathbf{\Sigma}_\alpha \mathbf{V}_\alpha^T, \tag{5}$$

identifies the independent rows of the primitive-coordinate differential space. Rows associated with singular values below the numerical threshold are discarded. The retained left-singular vectors define a row-orthonormal transformation

$$\mathbf{R}_\alpha = \mathbf{U}_{\alpha,r}^T, \tag{6}$$

where the subscript r denotes the retained singular subspace. The non-redundant coordinates for family α are then

$$\Delta\mathbf{q}_\alpha = \mathbf{R}_\alpha \Delta\mathbf{s}_\alpha, \qquad \mathbf{B}_{q_\alpha} = \mathbf{R}_\alpha \mathbf{B}_{s_\alpha}. \tag{7}$$

The number of rows of $\mathbf{R}_\alpha$ is therefore the rank of $\mathbf{B}_{s_\alpha}$, not an empirical number of coordinates chosen by the user. Stretching primitives are a special but important limiting case: when their block is already full rank, the transformation reduces to the identity and the primitive stretches are retained as independent GICs. For bending, torsional, out-of-plane, and ring blocks, the transformation removes only linear dependencies inside the same physical family.

Stacking the blocks gives

$$\Delta\mathbf{q} = \mathbf{R}\Delta\mathbf{s}, \qquad \mathbf{B}_q = \mathbf{R}\mathbf{B}_s. \tag{8}$$

The block-diagonal structure of $\mathbf{R}$ is the mathematical origin of the chemical homogeneity of the final coordinates: stretching information cannot replace a torsional or out-of-plane degree of freedom during rank reduction. After the family-wise reduction, the stacked matrix $\mathbf{B}_q$ is checked again against the expected vibrational rank. The global rank test is performed on $\mathbf{B}_q$, not on the coordinate labels. If residual dependencies remain because two chemically distinct families describe the same infinitesimal motion, the affected rows are identified by a rank-revealing decomposition of $\mathbf{B}_q$ with the family blocks kept explicit. The final pruning therefore removes only dependent rows inside the affected family blocks and never replaces a stretch by a bend or a torsion by an out-of-plane coordinate. If more than one row removal is numerically equivalent, deterministic tie breaking follows the same topological ordering. Only after this non-redundant unsymmetrized GIC basis has been fixed is molecular symmetry applied.

Symmetry is then applied to the non-redundant GIC space. For each operation g of the detected point group, the coordinate engine constructs the induced operation $\mathbf{D}_q(g)$ on $\Delta\mathbf{q}$ from the atom permutation, Cartesian rotation, and primitive-coordinate transformation.

The projector onto irreducible representation Γ is

$$\mathbf{P}_{\Gamma}^{q} = \frac{d_{\Gamma}}{|G|} \sum_{g \in G} \chi_{\Gamma}(g) \mathbf{D}_{q}(g), \tag{9}$$

where d_{Γ} and χ_{Γ} are the dimension and character of Γ. Orthonormal columns of $\mathbf{S}_{\Gamma}$ span the range of this projector, and the symmetry-adapted GICs are

$$\Delta\mathbf{Q}_{\Gamma} = \mathbf{S}_{\Gamma}^{T} \Delta\mathbf{q}. \tag{10}$$

Only the totally symmetric block is active in an equilibrium refinement,

$$\Delta\mathbf{Q} \equiv \Delta\mathbf{Q}_{A_1} = \mathbf{S}_{A_1}^{T} \mathbf{R} \Delta\mathbf{s} = \mathbf{B}_{Q} \Delta\mathbf{x}, \tag{11}$$

with

$$\mathbf{B}_{Q} = \mathbf{S}_{A_1}^{T} \mathbf{R} \mathbf{B}_{s}. \tag{12}$$

Cartesian displacements associated with GIC corrections are obtained from the rank-controlled minimum-norm solution

$$\Delta\mathbf{x} = \mathbf{B}_{Q}^{+} \Delta\mathbf{Q}, \tag{13}$$

where the Moore–Penrose pseudoinverse is evaluated with explicit singular value thresholds and topology validation after accepted steps.

When the task is not least-squares refinement but a geometry correction from target internal coordinates, the same derivative machinery can be used in weighted form. The Cartesian correction is obtained by minimizing

$$\left\| \mathbf{W}_{s}^{1/2} \left(\mathbf{B}_{t} \Delta\mathbf{x} - \Delta\mathbf{t} \right) \right\|^{2}, \tag{14}$$

where $\mathbf{t}$ contains requested internal-coordinate changes or auxiliary reference targets. The diagonal metric $\mathbf{W}_s$ gives different stiffness to different coordinate families, so the Cartesian back-transformation distributes corrections according to a chosen physical metric. Auxiliary targets are not part of the covalent GIC active space unless a dedicated coordinate block is explicitly defined.

These equations define how GIC corrections are converted into Cartesian geometry updates. The covariance propagation from the fitted active variables to Cartesian coordinates and finally to reported structural parameters is described below together with the treatment of constraints, because the same formalism is used for both the GIC and Cartesian active spaces.

2.5 Symmetry-Adapted Cartesian Coordinates

Although the default `SEfit` representation is the non-redundant symmetry-adapted GIC space described above, the same least-squares machinery can also operate in a symmetry-adapted Cartesian displacement basis. This provides a direct Cartesian route for geometry updates while using the same symmetry selection, constraint projection, and covariance propagation as the GIC route.

Starting from the reference Cartesian geometry, the first step is the removal of external translations and rotations. Let $\mathbf{E} \in \mathbb{R}^{3N \times n_{\text{ext}}}$ collect the external displacement vectors. For a nonlinear molecule $n_{\text{ext}} = 6$; for a linear molecule the rotational vector along the molecular axis is omitted. The three translational columns contain uniform displacements of all atoms along the laboratory axes. The rotational columns are generated from infinitesimal rotations around the center of mass,

$$\Delta \mathbf{r}_i = \boldsymbol{\omega} \times (\mathbf{r}_i - \mathbf{r}_{\text{cm}}) \,, \tag{15}$$

for each atom i. The columns of $\mathbf{E}$ are orthonormalized to give $\mathbf{Z}_{\text{ext}}$, and the internal

Cartesian projector is

$$\mathbf{P}_{\text{vib}} = \mathbf{I}_{3N} - \mathbf{Z}_{\text{ext}}\mathbf{Z}_{\text{ext}}^T. \tag{16}$$

This projector defines the non-redundant Cartesian displacement space before any symmetry classification. It is the Cartesian analogue of removing redundancies from the primitive internal-coordinate space: the resulting rank is $3N-6$ for nonlinear molecules and $3N-5$ for linear molecules.

For each irreducible representation Γ of the automatically detected point group, a Cartesian symmetry projector is constructed directly in the $3N$-dimensional displacement space as

$$\mathbf{P}_\Gamma^x = \frac{d_\Gamma}{|G|}\sum_{g\in G}\chi_\Gamma(g)\mathbf{R}(g), \tag{17}$$

where $\mathbf{R}(g)$ is the Cartesian operation, including atom permutation, associated with the symmetry operation g. Orthonormal columns spanning $\mathbf{P}_{\text{vib}}\mathbf{P}_\Gamma^x\mathbf{P}_{\text{vib}}$ define the symmetry-adapted Cartesian basis,

$$\text{range}\left(\mathbf{P}_{\text{vib}}\mathbf{P}_\Gamma^x\mathbf{P}_{\text{vib}}\right) = \text{span}\left(\mathbf{C}_\Gamma\right). \tag{18}$$

The columns of $\mathbf{C}_\Gamma$ are orthonormal Cartesian displacement directions. They are not normal modes and do not depend on a force field; they are only a symmetry-adapted basis of the translation- and rotation-free Cartesian space. In the refinement only the totally symmetric block is used,

$$\Delta\mathbf{x} = \mathbf{C}_{A_1}\Delta\mathbf{q}_{\text{SC}}, \tag{19}$$

where $\Delta\mathbf{q}_{\text{SC}}$ are amplitudes along the totally symmetric Cartesian directions. Primitive variations induced by these amplitudes are

$$\Delta \mathbf{s} = \mathbf{B}_s \mathbf{C}_{A_1} \Delta \mathbf{q}_{\mathrm{SC}}. \tag{20}$$

This equation is the link between the Cartesian active space and the same primitive-coordinate interface used by the GIC route. Primitive constraints, such as Gaussian-style `ModRedundant` freezes or local hydrogen constraints, are still specified in terms of internal coordinates and are projected onto the symmetry-Cartesian active space by the analytic derivative matrix $\mathbf{B}_s \mathbf{C}_{A_1}$. Direct Cartesian constraints would be less useful: a fixed bond or angle is a nonlinear, orientation-free internal relation, whereas its Cartesian representation is distributed over several coordinates and changes with the molecular frame. The final structural table is generated in the same way as for the GIC refinement: the optimized Cartesian geometry is used to evaluate ordinary bond lengths, valence angles, dihedral angles, and propagated uncertainties. Thus the working coordinates need not be chemical coordinates themselves; chemical interpretability is recovered in the reported primitive-coordinate layer.

2.6 Constraints, Predicates, and Parameter Classes

A key design principle of the coordinate layer is the separation between the chemical language used by the user and the active coordinate representation used by the numerical solver. User-defined structural relations are specified in terms of primitive internal coordinates,

$$\mathbf{s} = (s_1, s_2, \ldots, s_m)^T, \tag{21}$$

not in terms of GIC amplitudes or Cartesian components. This is essential because the chemically meaningful objects are bonds, valence angles, torsions, out-of-plane coordinates, ring primitives, and local hydrogen frames. A constraint such as a fixed C–H distance or an equality between two valence angles is local and orientation independent in primitive internal coordinates, whereas its Cartesian expression is nonlinear, distributed over several atoms, and dependent on the current molecular frame.

The same primitive relation can be projected onto either active coordinate space. Let $\Delta\mathbf{a}$ denote the active correction vector. For a GIC refinement, $\Delta\mathbf{a} = \Delta\mathbf{Q}$ and

$$\Delta\mathbf{x} = \mathbf{T}_A \Delta\mathbf{a}, \qquad \mathbf{T}_A = \mathbf{B}_Q^+. \tag{22}$$

For a Cartesian refinement, $\Delta\mathbf{a} = \Delta\mathbf{q}_{\mathrm{SC}}$ and

$$\Delta\mathbf{x} = \mathbf{T}_A \Delta\mathbf{a}, \qquad \mathbf{T}_A = \mathbf{C}_{A_1}. \tag{23}$$

Thus primitive corrections induced by any active coordinate model are

$$\Delta\mathbf{s} = \mathbf{B}_s \mathbf{T}_A \Delta\mathbf{a}. \tag{24}$$

This equation is the central projector for constraints, predicates, parameter classes, and error propagation.

Hard constraints. Hard constraints are exact relations imposed throughout the refinement. They may be individual freezes,

$$s_i = s_i^0, \tag{25}$$

symmetry or chemical equalities,

$$s_i - s_j = 0, \tag{26}$$

or more general linearized functions of primitive coordinates,

$$\mathbf{c}(\mathbf{s}) = \mathbf{d}. \tag{27}$$

At a trial geometry the corresponding first-order condition is

$$\mathbf{C}_s\Delta\mathbf{s} = \mathbf{b}_c, \qquad \mathbf{C}_s = \left.\frac{\partial \mathbf{c}}{\partial \mathbf{s}}\right|_{\mathbf{s}_0}, \qquad \mathbf{b}_c = \mathbf{d} - \mathbf{c}(\mathbf{s}_0). \tag{28}$$

Projection to the active space gives

$$\mathbf{C}_A\Delta\mathbf{a} = \mathbf{b}_c, \qquad \mathbf{C}_A = \mathbf{C}_s\mathbf{B}_s\mathbf{T}_A. \tag{29}$$

The solver does not remove every GIC or Cartesian direction that contains a constrained primitive. That would be basis dependent and could delete unrelated chemically valid motions. Instead, the allowed correction is written as

$$\Delta\mathbf{a} = \Delta\mathbf{a}_0 + \mathbf{N}\Delta\mathbf{p}, \qquad \mathbf{C}_A\mathbf{N} = \mathbf{0}, \tag{30}$$

where $\Delta\mathbf{a}_0$ is a minimum-norm particular solution of the linearized constraint equations and $\mathbf{N}$ spans their null space. The least-squares problem is solved only for the independent variables $\Delta\mathbf{p}$. In this way the forbidden component of the displacement is removed while all compatible directions remain available to the refinement.

When a primitive coordinate is constrained, all symmetry-equivalent primitives belonging to the same orbit are constrained before the null-space projector is formed. This prevents a local constraint, for example on one C–H distance, from breaking the molecular symmetry of the refinement. Gaussian-style `ModRedundant` freeze records are interpreted in this primitive layer and then completed over the relevant symmetry orbits. More general Gaussian-style generalized-internal-coordinate expressions are also admitted as hard constraints: arithmetic functions of primitive `R/A/D/U/L` coordinates or of final `GIC`n coordinates are evaluated at each trial geometry and projected through the same linearized constraint equations. Thus equality constraints, fixed sums or differences, and simple nonlinear functions are treated without deleting all coordinates that contain one constrained primitive.

The implemented syntax follows the Gaussian generalized-internal-coordinate style. A line `Name=Expression` defines a reusable coordinate, whereas the same named coordinate supplied with `Frozen` and `Value=` imposes a hard equation. Defined names can be used inside subsequent expressions, so chemically transparent constraints can be written as combinations of named stretches, angles, dihedrals, out-of-plane coordinates, Cartesian components, or final GIC labels. For angular-only expressions, a bare `Value=` target is interpreted in degrees, as in the Gaussian-style input convention used here; explicit `rad` targets are also accepted. If `Value=` is omitted, the current starting-geometry value is frozen. Tabulated structural information therefore uses `Value=` to keep the target independent of the starting Cartesian structure.

This mechanism is central for reduced-dimensionality refinements. In many rotational data sets, carbon-bound deuterated isotopologues are incomplete or absent, whereas deuteration of labile N–H or O–H groups may be available. The default strategy is therefore not to freeze all hydrogens blindly. Hydrogen coordinates supported by isotopic data remain refinable, while unsupported local hydrogen frames can be generated deterministically from the reference geometry and frozen by the same projection. The heavy-atom skeleton is then optimized without over-constraining chemically unrelated motions. Any additional model-dependent geometry information is handled as a reference relation and is kept outside the covalent topology used for GIC generation.

Predicate observations. Predicates are soft structural information, usually obtained from quantum-chemical calculations. Unlike hard constraints, they do not remove a degree of freedom; they add weighted residuals.[10] A primitive predicate has the form

$$f(\mathbf{s}) = f^{\mathrm{ref}} \pm \sigma_f, \tag{31}$$

and contributes

$$r_f = \frac{f(\mathbf{s}) - f^{\mathrm{ref}}}{\sigma_f} \tag{32}$$

to the residual vector. Its Jacobian with respect to the independent least-squares variables is

$$\mathbf{J}_f = \frac{1}{\sigma_f}\frac{\partial f}{\partial \mathbf{s}}\mathbf{B}_s\mathbf{T}_A\mathbf{N}. \tag{33}$$

Predicates are therefore treated on the same statistical footing as rotational observables, with their uncertainties controlling their influence.

Parameter classes. Parameter classes express the statement that several primitive quantities should be corrected together. Typical examples are C–H bond lengths, hydrogen-containing valence angles, methyl-group parameters, or aromatic C–C distances when the isotopic data are insufficient to determine every member independently. If class c contains primitives $i \in c$, the allowed primitive correction has the form

$$\Delta s_i = \Delta p_c, \qquad i \in c. \tag{34}$$

In practice, class definitions are converted into linear relations in the same primitive-coordinate layer as hard constraints, or into a compression of the Jacobian before the normal equations are solved. After convergence, the covariance is expanded back to the reported primitive and topological parameters. Thus classes reduce dimensionality without hiding which local structural quantities have been tied together.

The class operation applies to corrections rather than to final values. Thus two C–H bonds, two X–Y–H angles, or two heavy-atom bond families may retain different computed reference values while sharing the same fitted offset. This is the coordinate-independent analogue of the reduced-dimensionality regressions used in MSR, but here the classes are defined on automatically generated primitive-coordinate orbits and projected onto either the GIC or symmetry-Cartesian active space.

2.7 Semiexperimental Observables

For each isotopologue s, the measured ground-state rotational constants are converted into equilibrium rotational constants by removing vibrational and, when necessary, electronic contributions,[3–6]

$$B_{\alpha,e}^{(s)} = B_{\alpha,0}^{(s)} - \Delta B_{\alpha,\mathrm{vib}}^{(s)} - \Delta B_{\alpha,\mathrm{elec}}^{(s)}, \qquad \alpha \in \{a, b, c\}. \tag{35}$$

The correction terms may originate from any computational protocol and are therefore independent of the coordinate engine. In the present implementation, vibrational and electronic corrections are supplied through the isotopologue data file.

This decoupling is intentional. Starting geometries and vibrational corrections may be generated by any suitable electronic-structure or composite protocol.[11] `SEfit` does not assume that these data were generated by a specific quantum-chemical program. It stores the correction convention, numerical uncertainties, isotopic masses, and provenance in the isotopologue file, then performs the structural refinement with the chosen automatic coordinate model.

Although rotational constants may be fitted directly, `SEfit` uses principal moments of inertia as the default observables,

$$I_{\alpha}^{(s)} = \frac{K}{B_{\alpha,e}^{(s)}}, \tag{36}$$

where K is the standard rotational conversion constant.

Moments of inertia give a smoother dependence on geometry than rotational constants and avoid reciprocal amplification of small constant variations. The least-squares problem is therefore generally better conditioned, especially when isotopologues with different mass distributions are fitted together.

Direct fitting of rotational constants remains available for diagnostics and comparisons. For planar molecules, however, only two principal rotational components are independent.

If the molecular plane is the ab plane,

$$I_c = I_a + I_b, \tag{37}$$

and the three rotational constants are related through the reciprocal transformation $B_\alpha = K/I_\alpha$. Fitting all three components would therefore duplicate one experimental constraint and overstate the amount of independent information. `SEfit` consequently builds the least-squares normal equations from one of the rotational-component pairs AB, AC, or BC only. With the default moment-of-inertia target, these choices correspond to the pairs (I_a, I_b), (I_a, I_c), and (I_b, I_c), respectively. The omitted component is still calculated and reported after convergence as a diagnostic of the consistency of the vibrational corrections, electronic corrections, and planarity assumption.

The pair is selected by a deterministic rule. First, planarity is recognized from the singular values of the centered Cartesian geometry; a molecule is treated as planar when the smallest singular value divided by the largest geometrical scale is below 10^{-3}. If the user does not request a specific pair, `SEfit` builds the linearized Jacobian for each admissible candidate, AB, AC, and BC, and retains the pair with the largest numerical rank. Among rank-equivalent choices, it minimizes the propagated instability of the omitted moment using the planar identity $I_c = I_a + I_b$. Thus AB omits I_c and predicts it as $I_a + I_b$, whereas AC and BC omit an in-plane moment and recover it from a difference, $I_b = I_c - I_a$ or $I_a = I_c - I_b$. The score is evaluated over all isotopologues from the relative inconsistency of the reconstructed omitted moment plus its propagated uncertainty; the root-mean-square (RMS) value of this quantity is minimized. The Jacobian condition number is used only as the final tie-breaker. If a planar calculation is forced to use all three components, the input is rejected because ABC would introduce a redundant equation.

The complete experimental data set consists of the selected observables for all isotopologues and their uncertainties. These values form the experimental part of the least-squares objective described below.

2.8 Least-Squares Refinement

The semiexperimental refinement is formulated as a weighted nonlinear least-squares problem. The active coordinate vector $\mathbf{a}$ is either the totally symmetric GIC vector or the totally symmetric Cartesian displacement amplitude vector. After hard constraints and parameter classes have been projected, the actual independent least-squares variables are collected in $\mathbf{p}$.

The residual vector is

$$\mathbf{r} = \mathbf{y}^{\mathrm{obs}} - \mathbf{y}^{\mathrm{calc}}(\mathbf{p}), \tag{38}$$

where $\mathbf{y}$ collects all experimental observables and predicate observations included in the refinement.

The objective function is

$$\chi^2 = \mathbf{r}^T \mathbf{W} \mathbf{r}, \tag{39}$$

where $\mathbf{W}$ is the diagonal weight matrix constructed from the inverse variances of the experimental and predicate observations. When robust fitting is requested, the experimental part of $\mathbf{W}$ is modified by an iteratively reweighted loss applied to complete isotopologue blocks rather than to individual rotational constants. Thus all selected components of a suspect isotopologue are attenuated together, whereas quantum-chemical predicates retain their assigned statistical weights.

The optimization is performed using a weighted Levenberg–Marquardt procedure,

$$\left(\mathbf{J}^T \mathbf{W} \mathbf{J} + \lambda \mathbf{I}\right) \Delta \mathbf{p} = \mathbf{J}^T \mathbf{W} \mathbf{r}, \tag{40}$$

where $\mathbf{J}$ is the Jacobian matrix of the observables with respect to the independent variables $\mathbf{p}$ and λ is the damping parameter. Equation (40) is not solved in the implementation by forming the normal equations. The actual step is obtained from the column-scaled augmented system

$$\begin{bmatrix} \mathbf{W}^{1/2}\mathbf{J}\mathbf{S} \\ \lambda^{1/2}\mathbf{I} \end{bmatrix} \Delta\mathbf{u} = \begin{bmatrix} \mathbf{W}^{1/2}\mathbf{r} \\ \mathbf{0} \end{bmatrix}, \qquad \Delta\mathbf{p} = \mathbf{S}\Delta\mathbf{u},$$

using rank-revealing singular-value-decomposition (SVD) and QR-factorization linear algebra.[18] The diagonal scaling matrix $\mathbf{S}$ is updated from the current weighted Jacobian column norms, on top of the chemical coordinate-family scaling. This prevents weakly observed variables or mixed units from controlling the trust-region step and avoids the condition-number squaring associated with explicit normal equations.

The Jacobian is evaluated analytically whenever possible. Independent coordinate corrections are transformed into active-coordinate corrections, then into Cartesian displacements through the projector $\mathbf{T}_A\mathbf{N}$ described above. This gives analytic derivatives of moments of inertia and rotational constants with respect to the variables that are actually fitted.

The solver uses a trust-region Levenberg–Marquardt strategy based on the predicted and actual reduction of the weighted objective function.[19] Trial line-search points are evaluated with a frozen coordinate model. Every accepted step is then validated by the coordinate-definition service against the original topology, point group, irreducible-representation counts, and coordinate-family counts. Steps that would change this GIC signature are rejected and the trust radius is reduced. The Cartesian–GIC projector is refreshed analytically when needed and otherwise updated by a secant correction, avoiding unnecessary coordinate rebuilds without accepting topology-changing steps. The same numerical kernel is mirrored in the Fortran77 backend: dynamic column scaling, rank-revealing QR solution of the augmented Levenberg–Marquardt system, and grouped robust weights are available without importing legacy Gaussian code.

Convergence requires both the parameter corrections and the objective-function reduction to fall below predefined thresholds. Trial steps that increase the objective are rejected automatically, and the damping parameter is adjusted adaptively. For validation, `SEfit` can perform exact leave-one-isotopologue-out refits. Each isotopologue is removed in turn, the geometry is refitted with the same coordinate and constraint model, and the omitted rotational constants are predicted from the resulting structure. These diagnostics are not part of the objective function; they identify outlier isotopologues and quantify how strongly individual substitutions control the fitted geometry. The same philosophy is used for numerical warnings: reduced rank, small singular values, strongly downweighted isotopologues, ill-conditioned planar component pairs, and unusually large propagated coordinate uncertainties are reported explicitly, but they do not redefine the coordinate model or the least-squares objective.

Following convergence, the covariance matrix in the independent fitted variables is obtained from

$$\mathbf{C}_p = \sigma_r^2 \left(\mathbf{J}^T \mathbf{W} \mathbf{J}\right)^{-1}, \tag{41}$$

where σ_r^2 is the reduced residual variance.

2.9 Error Propagation

Uncertainty propagation follows the same chain of transformations used for the geometry update. If $\mathbf{N}$ is the null-space basis of the hard constraints and class reductions, the covariance in the active coordinate space is

$$\mathbf{C}_a = \mathbf{N} \mathbf{C}_p \mathbf{N}^T. \tag{42}$$

The corresponding Cartesian covariance is

$$\mathbf{C}_x = \mathbf{T}_A \mathbf{C}_a \mathbf{T}_A^T. \tag{43}$$

For a GIC refinement, $\mathbf{T}_A = \mathbf{B}_Q^+$; for a Cartesian refinement, $\mathbf{T}_A = \mathbf{C}_{A_1}$. Therefore the same covariance propagation is used for both coordinate choices.

Any reported structural quantity is treated as a function of the optimized Cartesian geometry. For a vector of reported quantities $\mathbf{g}(\mathbf{x})$, which may contain bond lengths, valence angles, dihedral angles, out-of-plane coordinates, ring descriptors, or user-defined primitive functions, the covariance is

$$\mathbf{C}_g = \mathbf{G}_x \mathbf{C}_x \mathbf{G}_x^T, \qquad \mathbf{G}_x = \left. \frac{\partial \mathbf{g}}{\partial \mathbf{x}} \right|_{\mathbf{x}_{\text{opt}}}. \tag{44}$$

The reported standard errors are the square roots of the diagonal elements of $\mathbf{C}_g$, while the off-diagonal elements provide correlations among derived structural parameters. Angular derivatives are evaluated in radians and converted to degrees only in the final report. Symmetry-equivalent reported quantities can either be listed separately with their covariance or condensed by the same linear propagation formula.

Hard constraints have no variance along the constrained directions because those directions are removed before the normal equations are solved. Predicate observations are different: their stated uncertainties enter the weight matrix and therefore contribute statistically to $\mathbf{C}_p$. This distinction is important when hydrogen parameters are held fixed because deuterated isotopologues are unavailable. A frozen local hydrogen frame is a constraint; a quantum-chemical C–H distance with an assigned uncertainty is a predicate. Only the latter contributes uncertainty through its assigned standard deviation.

The output therefore reports both standard errors and correlation diagnostics. Large correlations are not treated as failures by themselves, because they are expected whenever isotope substitutions do not isolate all local structural directions. They are used to identify underdetermined parameter combinations, to decide whether a primitive should be fixed or converted into a predicate, and to verify that reported derived quantities are propagated from the full covariance matrix rather than from independent one-parameter estimates. The same information is complemented by an SVD diagnostic of the final weighted Jacobian. Small singular values are reported together with the dominant coordinate combinations in the corresponding right singular vectors, so a rank deficiency can be traced to specific GICs, parameter classes, or symmetry-Cartesian directions instead of being summarized only by a condition number.

All constraints are also audited after projection. The output records the input primitive patterns, the symmetry-equivalent primitive constraints generated from them, the parameter classes, and the active working-coordinate labels actually matched by each entry. This is important for black-box refinements: the user does not construct the active coordinate basis manually, but can verify exactly which chemically stated restrictions were imposed on that basis.

The final structural tables are thus independent of the internal coordinate basis used during optimization. The fitted variables may be GIC amplitudes or Cartesian amplitudes, but bond lengths, angles, dihedrals, and ring parameters are always evaluated from the same optimized Cartesian geometry and receive errors from the same propagated covariance matrix.

After convergence, `SEfit` also evaluates the numerical Hessian of the weighted least-squares objective with respect to the independent variables that remain after symmetry selection, parameter-class reduction, and constraint projection. Diagonalization of this optimized-space Hessian provides the final stationary-point diagnostic: a positive spectrum identifies a local minimum in the fitted coordinate space, negative eigenvalues flag an unstable direction, and near-zero eigenvalues are reported as rank-deficient or statistically flat directions rather than hidden by the refinement. The benchmark tables therefore report the stationary-point classification, and, where useful, the smallest Hessian eigenvalue.

2.10 Kraitchman Diagnostic

For singly substituted isotopologues, `SEfit` computes substitution coordinates from the Kraitchman equations[20] and compares their absolute values with the fitted coordinates in the parent principal-axis frame. This diagnostic is intentionally separate from the fit: Kraitchman coordinates are useful assignment checks, but they do not retain coordinate signs and do not exploit the full correlated least-squares model.

3 Implementation and Reproducibility

The implementation is organized around two reusable services. The coordinate-definition service defines the molecular coordinate model from Cartesian atoms and nuclear charges, assigning point-group symmetry and irreducible representations when requested. A separate B-matrix service evaluates the same frozen GIC definition for any compatible Cartesian geometry. `SEfit` calls the definition service only at a fresh start and reuses the stored coordinate model during the least-squares iterations, rebuilding or updating the B matrix only when the current step requires it. The command-line interface and the graphical interface call these same services and expose the choice between GIC and symmetry-adapted Cartesian active spaces.

The standard input model contains Cartesian parent geometry, optional Gaussian-style primitive freeze records, and a separate isotopologue file with substitutions, ground-state rotational constants, vibrational and electronic corrections, uncertainties, and correction conventions. Legacy MSR-style input is accepted for reproducibility, including frozen variables marked by `#`, but it is converted immediately into the same Cartesian, primitive-constraint, and isotopologue representation. Z-matrices are therefore compatibility inputs, not working coordinates.

Each run writes a text report, final Cartesian coordinates, primitive internal parameters with propagated uncertainties, corrected/calculated rotational observables, residuals, covariance/correlation and influence diagnostics, SVD, constraint, and warning diagnostics, Hessian eigenvalues for the minimum check, Kraitchman diagnostics when applicable, and a machine-readable manifest. A checkpoint file stores the Cartesian geometry, active labels, damping/trust region state, and robust weights; restart calculations rebuild the GIC or symmetry-Cartesian definitions deterministically from those Cartesian coordinates. The benchmark tables in this article are generated from the versioned run comma-separated-value (CSV) files by a single regression command, so the manuscript values, test suite, and archived examples use the same numerical source. A versioned distribution of the `SEfit` source code, examples, and regression tests will be made available to reviewers and, upon publication, under the distribution model selected by the authors.

4 Comparison with Existing Approaches

The present methodology combines ideas from traditional internal-coordinate representations, generalized internal coordinates, and semiexperimental structure refinement. Its distinguishing feature is that the coordinate model is generated once from molecular topology and then reused independently of the application backend.

4.1 Comparison with Z-Matrix-Based Approaches

Z-matrices remain useful when the molecular graph is essentially tree-like. For glycolaldehyde, for example, a conventional Z-matrix can be written with little ambiguity and provides a compact set of recognizable parameters. In such a case the non-uniqueness of the reference tree has little practical effect because the chemically important distances and angles can all be made primary variables.

The limitations appear when the same strategy is applied to rings, fused systems, and bridged frameworks. A tree cannot represent all ring distances as primary coordinates at once; at least one ring-closing distance is necessarily derived. The selected atom order can also make symmetry-equivalent fragments appear different unless the user imposes additional relations. This is not a minor formatting issue. In a least-squares refinement, the covariance matrix is defined for the variables that are actually fitted. A bond that is primary in one Z-matrix and derived in another receives its uncertainty through a different Jacobian, and correlated ring quantities may not be represented uniformly unless the user explicitly designs the model to do so.

`SEfit` avoids this asymmetry. The input is a Cartesian geometry, topology is perceived automatically, and all primitive internal coordinates are generated before reduction or symmetry adaptation. The GIC and symmetry-Cartesian routes then provide two non-redundant active spaces that do not privilege one traversal of the molecule over another. This does not remove chemical constraints; it moves them to the primitive-coordinate interface, where they are chemically meaningful and can be projected consistently onto either active basis. All reported structural parameters are then evaluated from the final Cartesian geometry with the same covariance propagation machinery, independently of whether the parameter was a fitted amplitude, a primitive coordinate, or a topological descriptor derived after the fit.

4.2 Comparison with Conventional Generalized Internal Coordinates

Generalized internal coordinates extend traditional internal-coordinate systems by allowing linear combinations of primitive structural variables. They are widely used in geometry optimizations, vibrational analyses, reaction-path calculations, and targeted exploration of interatomic geometries.[12,14–17,21]

Most implementations still require the user to define generalized coordinates or coordinate transformations explicitly. The present workflow changes the division of labor: primitive coordinates are the user-visible chemical layer, whereas generalized coordinates are an internal, automatically generated representation.

4.3 Comparison with Existing GIC Implementations

Programs such as Gaussian provide powerful generalized-internal-coordinate formalisms for coordinate combinations, constraints, scans, and conditional coordinate definitions. The recent Gaussian GIC implementation of Marenich, Brothers, Hratchian, and Frisch makes

GICs a flexible language for exploring interatomic geometries.[15,21]

The present workflow uses generalized coordinates differently. In Gaussian-style workflows, GICs remain a user-defined, generally redundant coordinate layer, and non-standard cyclic, fused, bridged, or symmetry-constrained cases still require the user to specify or curate the coordinates. Here, generalized coordinates are internal variables derived from topology-based primitives, reduced to a non-redundant space, assigned to irreducible representations, and accessed through primitive-coordinate constraints, predicates, and classes. This separation reduces coordinate engineering, especially for cyclic, fused-ring, and bridged systems.

4.4 Comparison with MSR, STRFIT, and Semiexperimental Refinement Programs

The semiexperimental refinement strategy implemented in `SEfit` is directly derived from the principles introduced in the MSR program,[7] including weighted nonlinear least-squares refinement, the use of moments of inertia as observables, predicate observations,[10] uncertainty propagation, leverage and correlation diagnostics, and final minimum checks. Recent MSR developments also added automated computed starting geometries, efficient vibrational corrections, and reduced-dimensionality treatments for incompletely substituted systems.[11] `SEfit` is intended to inherit this complete functionality, not merely to reproduce a subset of MSR fits.

The primary distinction lies in the coordinate and user-interface infrastructure on which the refinement is based. MSR and related programs normally rely on user-defined internal-coordinate models, typically based on Z-matrix-like structural parameters and manually selected relationships among them. In contrast, `SEfit` operates entirely within the coordinate framework generated by the coordinate engine: the input structure is Cartesian, no Z-matrix is constructed, and the active refinement variables are supplied by the common coordinate engine. Reduced-dimensionality choices, such as freezing unsupported C–H local frames to computed reference values or tying heavy-atom classes by common offsets, are expressed as primitive-coordinate relations and then projected automatically.

This design preserves the statistical strengths of established semiexperimental methodologies while shifting the user effort from coordinate engineering to chemically transparent constraints, isotopologue data, correction provenance, and diagnostic interpretation.

Another relevant point of comparison is Kisiel's `STRFIT`, distributed through the PROSPE package.[8] `STRFIT` is a general structure-fitting program widely used in rotational-spectroscopy studies to fit structural parameters to moments of inertia or related rotational observables. Recent semiexperimental applications use `STRFIT` with equilibrium rotational constants corrected by computed vibrational effects, but the structural model remains user supplied; in a representative cyclic case, the published workflow used Z-matrix input, kept hydrogen-containing parameters at computed values, and treated some ring quantities as derived Cartesian quantities outside the fitted covariance matrix.[9]

Overall, `SEfit` keeps the statistical machinery of established semiexperimental refinement while removing the need to design a molecule-specific Z-matrix or symmetry-coordinate model. The practical advantage is modest for standard acyclic systems and becomes decisive for cyclic, fused, bridged, or heavily constrained molecules.

5 Validation and Representative Applications

The validation set was chosen to separate standard and non-standard coordinate problems. Glycolaldehyde is a small acyclic molecule for which a traditional Z-matrix is straightforward; it verifies that the automatic procedure does not lose efficiency in a simple case. Glycine II tests a reduced-dimensionality nonplanar refinement in which experimental isotopic information is insufficient without chemically defined functional constraints. Cyclopentadiene, nitrobenzene, azulene, and norcamphor are more demanding because rings, planar rotational redundancy, fused topology, bridged connectivity, and missing deuterium substitutions force

Table 1: Operational comparison between representative semiexperimental or rotational-structure refinement workflows. The comparison concerns workflow automation and coordinate handling, not the statistical quality of individual refinements. MSR provides advanced semiexperimental least-squares machinery and recent automated starting-geometry/vibrational-correction support; `STRFIT`/PROSPE is a widely used structure-fitting route in rotational spectroscopy; `SEfit` retains these semiexperimental capabilities while shifting the coordinate model to an automatically generated topology-driven layer.

Feature	MSR workflow	`STRFIT`/PROSPE workflow	`SEfit` workflow
Primary geometry model	User-defined Z-matrix/internal-coordinate model	User-defined structural model, commonly Z-matrix or Cartesian-derived input	Cartesian geometry with automatic topology perception
Starting geometry generation	External or automated computed geometry in recent workflows	External to the structure-fit program	Computed or user Cartesian geometry with stored provenance
Vibrational/electronic corrections	Supported; recent workflows automate computed corrections	Supplied externally	Supplied as isotopologue data, independent of the coordinate engine
Manual Z-matrix/reference-tree dependence	Present in standard use	Present in many structural-fit applications	Not used
Manual symmetry-coordinate definition	Required or user-guided	Not an automatic active-GIC symmetry layer	Automatic from detected point group
Ring and fused-ring treatment	Manual coordinate design	Manual parameter selection; some ring quantities may be derived after the fit	Automatic ring primitives and ring GICs
Redundancy handling	Avoided by manual coordinate selection	Avoided by user-selected fitted parameters	Automatic rank reduction of primitive GIC space
Active refinement variables	User-selected structural parameters	User-selected structural parameters fitted to moments or constants	Automatically selected totally symmetric GICs or symmetry Cartesians
Parameter classes and predicates	Supported	Fixed/selected parameters are available through the input model	Supported and projected from primitive coordinates
Primitive-coordinate constraints	User-defined in the selected model	Fixed parameters in the supplied structural model	Projected automatically onto either active space
Missing C–H deuteration	Reduced-dimensionality models or predicates in the chosen coordinate model	Fixed parameters in the supplied model	Symmetry-complete local frames, hard constraints, predicates, or classes
Auxiliary geometry corrections	Included through the selected starting-geometry model	External to the fit model	Reference targets excluded from covalent GIC topology
Symmetry completion of constraints	User responsibility	User responsibility	Automatic over symmetry-equivalent primitive orbits
Uncertainties for derived topology	Model-dependent	Limited when quantities are derived outside the fitted covariance matrix	From optimized Cartesian structure with covariance propagation
Rank/outlier diagnostics	Leverage and correlation diagnostics	Program- and workflow-dependent	SVD coordinate-combination diagnostics, robust isotopologue weights, and optional leave-one-out refits
Software availability	Research code described in the literature	Downloadable from the PROSPE site	Versioned distribution planned with examples and regression tests

conventional workflows to choose derived coordinates, fixed hydrogen models, or manually coupled parameters.

No molecule-specific coordinate code was introduced. In every benchmark the input structure was Cartesian. The coordinate engine generated the coordinate model, `SEfit` selected the totally symmetric active space, and any constraints were supplied as primitive internal-coordinate relations. The same constraint definitions were used when the active basis was changed from GICs to symmetry-adapted Cartesians.

The regression suite additionally checks GIC generation for pyridine, uracil, guanine, water, cyclopentadiene, nitrobenzene, and azulene. These tests verify point-group assignment, non-redundant coordinate counts, and totally symmetric dimensions independently of the semiexperimental fit.

The atom numbering used in structural tables and diagnostic discussions is shown in Figure 2. The labels come directly from the Cartesian input order and the perceived topology, so they match the numbering used by the coordinate engine, `SEfit`, and the primitive-coordinate constraints.

Table 2: One-page summary of the representative `GIC_Forge/SEfit` semiexperimental refinements. "Model" is the working-coordinate model used for the main row. "Fit" gives the independent rotational components entering the normal equations; for moment-based fits, AB denotes (I_a, I_b). "GIC/A" gives the final non-redundant GIC count and the totally symmetric active count before constraint projection. "Eff./rank" gives the independent variables after primitive-constraint projection and the final numerical rank. Rotational residuals are RMS/maximum absolute values in MHz over all reported diagnostic constants.

System	PG	Model	Fit	Iso.	GIC/A	Eff./rank	Constraints	Steps	RMS/max
Glycolaldehyde	C_s	GIC	ABC	10	18/12	12/12	none	3/0; min	0.5640/1.7961
Glycine II	C_1	GIC	ABC	10	24/24	19/19	5 function	16/30; min	0.2491/0.6707
Cyclopentadiene	C_{2v}	Cart.	ABC	14	27/10	10/10	none	10/0; min	0.0635/0.2289
Nitrobenzene	C_{2v}	GIC	AB	9	36/13	13/13	8 primitive	4/0; pos.	0.1480/0.2605
Azulene	C_{2v}	GIC	AB	12	48/17	10/10	8 primitive	8/28; min	0.0731/0.2900
Norcamphor	C_1	GIC	ABC	9	48/48	18/18	30 H primitive	8/0; min	0.0710/0.2091

Table 3 makes the planar choice explicit. The selected AB pair is not an arbitrary convention: it retains the two in-plane moments directly and leaves the perpendicular constant as a diagnostic. Forced AC or BC fits have the same rank but give larger complete rota-

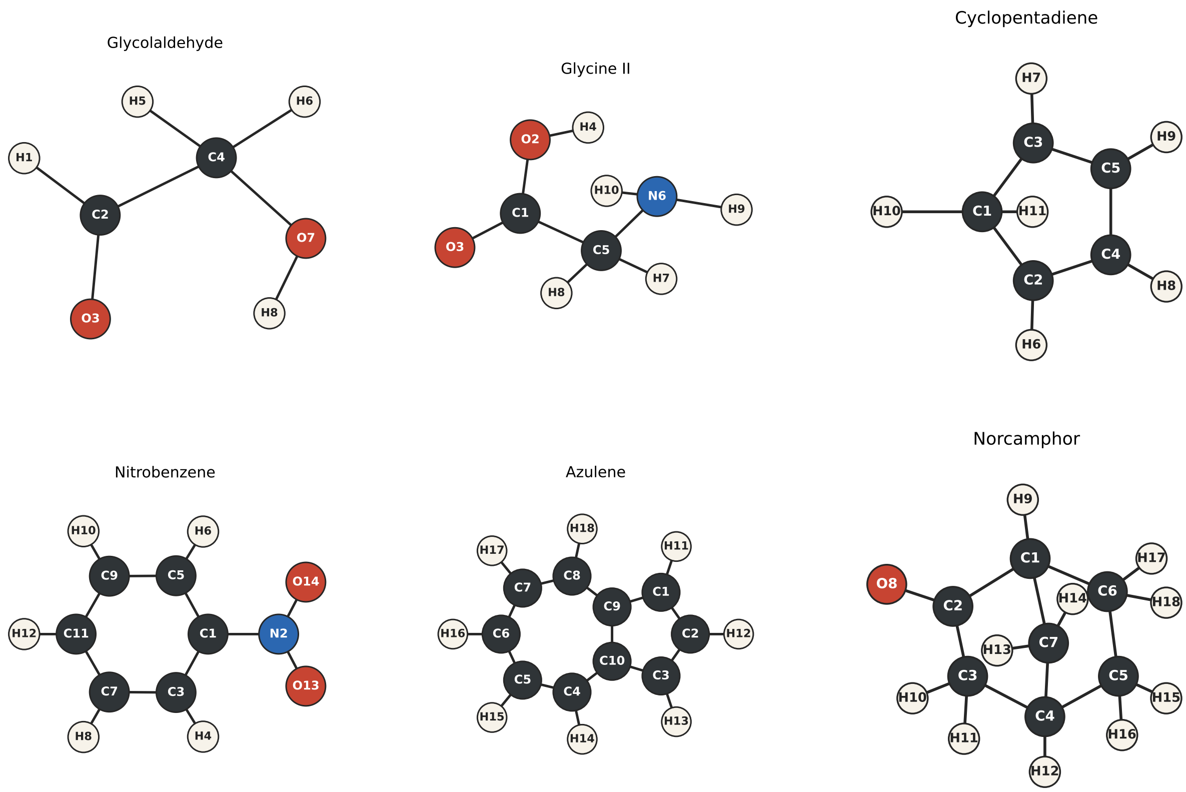

Figure 2: Atom numbering used for the glycolaldehyde, glycine II, cyclopentadiene, nitrobenzene, azulene, and norcamphor semiexperimental refinements. The labels are derived from the Cartesian input order and are used consistently in the GIC definitions, primitive-coordinate constraints, and structural tables.

Table 3: Planar-component diagnostics for the planar benchmarks. Only two principal moments are independent for a planar molecule; the selected pair marked by an asterisk is used in the normal equations, while all three rotational constants are recomputed after convergence. The residuals are RMS/maximum absolute MHz values over the complete diagnostic A, B, and C constant set.

System	Pair	Fitted moments	Rank	Cond.	Diagnostic RMS/max
Nitrobenzene	AB*	(I_a, I_b)	13	27464.4	0.1480/0.2605
Nitrobenzene	AC	(I_a, I_c)	13	56007.7	0.2593/0.4597
Nitrobenzene	BC	(I_b, I_c)	13	10623.3	2.4602/4.3693
Azulene	AB*	(I_a, I_b)	10	956.7	0.0731/0.2900
Azulene	AC	(I_a, I_c)	10	1780.0	0.1086/0.4222
Azulene	BC	(I_b, I_c)	10	950.2	0.3232/0.7541

tional diagnostic residuals for both nitrobenzene and azulene. In Table 2, "pos." denotes a rank-deficient statistical Hessian with positive curvature in all retained fitted directions.

5.1 Coordinate Content of the Benchmark Refinements

The least-squares variables are not necessarily the topological quantities listed in the final structural tables. They are either symmetry-adapted GICs or symmetry-adapted Cartesian amplitudes. In both cases, reported bond lengths, angles, dihedrals, and out-of-plane quantities are evaluated after convergence from the optimized Cartesian geometry, with uncertainties propagated from the fitted covariance matrix.

The primitive layer contains the local coordinates implied by the perceived graph: stretches, valence angles, torsions, and ring deformation coordinates derived from endocyclic valence-angle and torsional primitives. This redundant set is reduced and symmetry-adapted in homogeneous blocks. Stretching GICs are therefore formed only from stretches, valence-angle GICs only from valence-angle primitives, and torsional or ring-puckering GICs only from torsional-type primitives. This prevents a bond-stretching correction from being replaced by a ring-puckering or torsional correction during rank reduction.

The coordinate counts in Table 4 refer to the covalent-topology GIC space. Auxiliary reference targets are deliberately not included in these counts, because they do not create additional ring or GIC primitives.

Table 4 summarizes the GIC coordinate content. For glycolaldehyde the coordinate problem is standard: the active A' variables are stretches and bends. Glycine II and norcamphor are complementary C_1 cases: all GICs are formally totally symmetric, so dimensionality reduction comes from exact chemical constraints rather than symmetry. For nitrobenzene and azulene, planar aromatic symmetry leaves stretching and in-plane bending coordinates in the totally symmetric active space, whereas torsional and out-of-plane ring coordinates are retained in the complete coordinate report but excluded from the equilibrium refinement.

Primitive constraints reduce the active spaces only after the totally symmetric coordinates

Table 4: Coordinate content of the benchmark refinements. Numbers in parentheses are the totally symmetric coordinates within each family. The active variables are the totally symmetric coordinates before any primitive-constraint projection; in C_1, all coordinates transform as A.

System	Total	Stretching	Bending	Torsion/ring	Active totally symmetric character
Glycolaldehyde	18	7 (6)	8 (6)	3 (0)	6 stretches + 6 bends
Glycine II	24	9 (9)	10 (10)	5 (5)	all GICs in C_1; 5 functional constraints project to 19 variables
Cyclopentadiene	27	11 (6)	10 (4)	6 (0)	6 stretches + 4 bends
Nitrobenzene	36	14 (8)	11 (5)	11 (0)	8 stretches + 5 bends
Azulene	48	19 (11)	14 (6)	15 (0)	11 stretches + 6 bends
Norcamphor	48	18 (18)	25 (25)	5 (5)	all GICs in C_1; 30 primitive H constraints project to 18 variables

have been selected. In azulene, selected MSR-inspired C–H distances and valence angles reduce the 17 totally symmetric GICs to ten effective variables. In glycine II, five Gaussian-style functional constraints couple C–H and N–H differences, methylene rocking/twisting combinations, and the average H–N–C–C torsion, reducing the 24-dimensional C_1 active space to 19 variables. In norcamphor, the absence of deuterium substitutions is handled by freezing one C–H stretch, one C–C–H angle, and one C–C–C–H torsion for each hydrogen, leaving an 18-dimensional heavy-skeleton refinement. The same chemical information is therefore used as in the published reduced-dimensionality models, but the least-squares variables are generated automatically rather than supplied as Z-matrix parameters.

5.2 Glycolaldehyde Small-Molecule Benchmark

Glycolaldehyde was included as a compact acyclic benchmark. It is not a case where Z-matrix methods are expected to fail; rather, it verifies that the automatic Cartesian/topology workflow reproduces a standard refinement without requiring any manual coordinate model. The input geometry was the published composite-scheme Cartesian structure reported in the Supporting Information of a recent rotational-spectroscopy benchmark, and the spectroscopic data consisted of the parent species, singly substituted ^{13}C and ^{18}O isotopologues, hydroxyl and carbon-bound deuterated species, and two doubly substituted deuterated species. The ground-state rotational constants were corrected with computed vibrational contributions using the additive convention $B_e = B_0 + \Delta B_{\mathrm{vib}}$.

The coordinate engine assigned C_s symmetry and generated 18 final non-redundant GICs: seven stretching, eight bending, and three torsional coordinates. Twelve transform as A' and were active in the refinement; the A'' coordinates were retained in the complete report but excluded from the equilibrium fit. The refinement converged in three accepted steps without rejections, and the optimized-space Hessian was positive definite.

Table 5 gives the diagnostics. The rotational-constant RMS residual is larger than for the MSR-derived cyclopentadiene benchmark because the absolute residuals are dominated by the large A constants, while the fitted moment-of-inertia residuals remain small. The largest rotational residual is 1.7961 MHz for the A component of the ^{13}C2 isotopologue. A hydrogen-local-frame frozen comparison also converges to a minimum but gives a larger RMS residual of 1.7893 MHz, showing that the deuterated data carry useful information on hydrogen positions. A sign check using subtractive vibrational corrections gives a much poorer RMS residual of 15.99 MHz, so the additive convention is retained.

Table 5: Glycolaldehyde small-molecule semiexperimental benchmark. Moments of inertia were used as fit observables; rotational residuals are reported in MHz.

Quantity	Value
Isotopologues	10
Rotational observables	30
Final non-redundant GICs	18
Active A' GICs	12
Numerical rank	12
Accepted/rejected steps	3/0
Condition number	347.82
Weighted RMS in inertia space	2.83×10^{-3}
RMS rotational-constant residual	0.5640
Maximum rotational-constant residual	1.7961
Reduced χ^2	1.34×10^{-5}
Smallest Hessian eigenvalue	4.3757
Stationary point	minimum

5.3 Glycine II Functional-Constraint Benchmark

Glycine conformer II is a reduced-dimensionality benchmark because the available isotopic information does not by itself determine a stable semiexperimental structure. Kasalová et al. showed that high-level all-electron cc-pVTZ CCSD(T) structural information and vibrational corrections are needed to obtain a meaningful Gly-IIn refinement from the microwave isotopologue set.[22] This makes Gly-II a useful test of whether model information can be supplied as chemically transparent constraints without reverting to a Z-matrix.

The present calculation starts from the non-planar DPCS3 Cartesian geometry and uses the 10-isotopologue Gly-IIn data set with additive zero-point vibrational corrections. No C_s constraint is imposed: GICForge assigns the C_1 coordinate space, so all 24 final GICs are formally totally symmetric. The five reduced-dimensionality relations have the same functional form as the constraints used in the published refinement: a C–H difference, an N–H difference, two methylene angular combinations, and an average H–N–C–C torsion. They are supplied in Gaussian-style syntax as named coordinate definitions followed by `Frozen,Value=` equations, for example `RCH7=R(5,7)` and `DRCH(Frozen,Value=0.000091749296)=RCH8-RCH7`. Thus the constraints are functions of primitive coordinates, not fitted Z-matrix variables. Their target values were evaluated from the DPCS3 starting geometry.

Table 6 compares the DPCS3-derived targets with a run using the published Kasalová-type target values. The two fits have the same rank and essentially the same residual scale; the DPCS3 targets therefore reproduce the coupled-cluster constraint information closely enough for this benchmark while remaining generated by the same upstream Cartesian-geometry machinery used elsewhere in the workflow.

Table 6: Glycine II constraint-source comparison. Both calculations used GIC coordinates, moments of inertia as observables, and the same 10 corrected Gly-IIn rotational data sets. Rotational residuals are reported over the diagnostic A, B, and C constants in MHz.

Constraint target source	Rank	RMS	Max	$10^3\langle\Delta B^2\rangle$
Kasalová-type values	19	0.2344	0.8100	54.94
DPCS3-derived values	19	0.2491	0.6707	62.06

The final DPCS3-constrained refinement converges to a minimum with 19 effective variables, a condition number of 1.05×10^4, and a complete rotational RMS residual of 0.2491 MHz. The larger uncertainties are associated with the amino-hydrogen frame, as expected from the incomplete substitution pattern. Table 7 reports the final topological parameters and propagated standard errors. These quantities are evaluated from the optimized Cartesian geometry; the fitted coordinates themselves are the constrained, non-redundant GIC amplitudes.

Table 7: Glycine II DPCS3-constrained SEfit structural parameters. Bond lengths are in Å; angles and dihedrals are in degrees. Standard errors are propagated from the fitted covariance matrix after projection of the five functional constraints.

Parameter	Description	Value	Std. error
R(1,2)	C–O(H)	1.33327	0.00273
R(1,3)	C=O	1.20471	0.00297
R(1,5)	C–C	1.52276	0.00223
R(2,4)	O–H	0.99029	0.00246
R(5,6)	C–N	1.45798	0.00354
R(5,7)	C–H7	1.08732	0.00139
R(5,8)	C–H8	1.08742	0.00139
R(6,9)	N–H9	1.01282	0.00540
R(6,10)	N–H10	1.01425	0.00540
A(2,1,3)	O–C–O	123.126	0.387
A(2,1,5)	O(H)–C–C	114.228	0.130
A(3,1,5)	O=C–C	122.085	0.245
A(1,2,4)	C–O–H	105.236	0.154
A(1,5,6)	C–C–N	111.940	0.193
A(7,5,8)	H–C–H	106.906	0.196
A(9,6,10)	H–N–H	107.699	0.808
D(5,1,2,4)	C–C–O–H	-177.692	2.257
D(2,1,5,6)	O(H)–C–C–N	-174.255	2.307
D(3,1,5,6)	O=C–C–N	14.088	0.740
D(1,5,6,9)	C–C–N–H9	39.622	1.819
D(1,5,6,10)	C–C–N–H10	-84.075	1.819

5.4 Cyclopentadiene MSR Benchmark

The cyclopentadiene refinement used Cartesian coordinates and isotopologue rotational data corresponding to the MSR benchmark. The generated coordinate set contained 27 non-redundant symmetry-assigned GICs, of which 10 transformed as A_1 and were active. During the trust-region refinement, accepted steps were checked against the original GIC signature so that the C_{2v} topology and the A_1 active space were preserved.

The least-squares problem used moments of inertia and all three principal components for 14 isotopologues. No predicates, parameter classes, or automatic pruning were required. The optimization converged in six accepted steps without rejections, and the optimized-space Hessian was positive definite. The diagnostics are summarized in Table 8.

Table 8: Cyclopentadiene semiexperimental refinement from the MSR-derived data set. Moments of inertia are in amu Å^2 and rotational residuals are in MHz.

Quantity	Value
Isotopologues	14
Rotational observables	42
Final non-redundant GICs	27
Active A_1 GICs	10
Numerical rank	10
Accepted/rejected steps	6/0
Condition number	454.54
Weighted RMS in inertia space	7.18×10^{-4}
Maximum inertia residual	2.61×10^{-3}
RMS rotational-constant residual	0.0635
Maximum rotational-constant residual	0.2289
Reduced χ^2	6.77×10^{-7}
Smallest Hessian eigenvalue	0.9788
Stationary point	minimum

The largest residual in the rotational-constant representation is 0.229 MHz, while the RMS residual over the 42 corrected rotational constants is 0.0635 MHz. In the fitted moment-of-inertia space, the largest residual is 2.61×10^{-3} amu Å^2, and the RMS residual is 7.18×10^{-4} amu Å^2. The complete table of corrected experimental rotational constants, calculated constants, and differences is reported in the Supporting Information.

The same 14-isotopologue MSR-like data set was also refined with the symmetry-Cartesian coordinate model. The Cartesian calculation used 27 vibrational displacement directions after removal of translations and rotations, of which 10 were totally symmetric and active. It converged to a minimum with the same numerical rank and essentially the same residuals as the GIC refinement, giving a condition number of 635.4 and RMS/maximum rotational-constant residuals of 0.0635/0.2289 MHz. The comparison with the corresponding

GIC result is included in Table 11.

Selected structural parameters are reported in Table 9. The uncertainties are propagated from the optimized A_1 GIC covariance matrix to topological bond lengths, valence angles, and dihedrals; symmetry-equivalent quantities are condensed.

Table 9: Selected final topological parameters for cyclopentadiene. Bond lengths are in Å, angles and dihedrals in degrees. Reported uncertainties are one standard deviation.

Parameter	Atoms	Value
R(C–C)	1–2 / 1–3	1.49921 ± 0.00029
R(C=C)	2–4 / 3–5	1.34618 ± 0.00031
R(C–C)	4–5	1.46538 ± 0.00042
R(C–H)	1–10 / 1–11	1.09446 ± 0.00015
R(C–H)	2–6 / 3–7	1.07881 ± 0.00019
R(C–H)	4–8 / 5–9	1.07893 ± 0.00016
A(C–C–C)	2–1–3	103.069 ± 0.028
A(C–C–C)	1–2–4 / 1–3–5	109.336 ± 0.020
A(C–C–C)	2–4–5 / 3–5–4	109.130 ± 0.014
A(H–C–H)	10–1–11	106.578 ± 0.021
A(C–C–H)	1–2–6 / 1–3–7	123.942 ± 0.077
A(C–C–H)	4–2–6 / 5–3–7	126.722 ± 0.076
A(C–C–H)	2–4–8 / 3–5–9	126.059 ± 0.044
A(C–C–H)	5–4–8 / 4–5–9	124.811 ± 0.036
D(C–C–C–C)	3–1–2–4	-179.9928 ± 0.0000014
D(C–C–C–C)	1–2–4–5	179.9952 ± 0.0000006
D(C–C–C–C)	2–4–5–3	180.0000 ± 0.00000001

5.5 Cyclic and Polycyclic Systems

The cyclopentadiene benchmark tests a situation where a Z-matrix can be written but cannot treat every ring bond and deformation on the same footing. The automatic GIC model uses endocyclic valence-angle and torsional primitives rather than Cremer–Pople variables or a ring-closing Z-matrix distance. The symmetry-Cartesian run reaches the same rank and residuals without using internal coordinates as active variables. The same principle is then applied to nitrobenzene, azulene, and norcamphor.

5.6 Nitrobenzene Planar-Observable Benchmark

Nitrobenzene tests two additional features: import of an MSR-style legacy input containing a Z-matrix and dummy atoms, and automatic selection of independent rotational information for a planar molecule. The calculation itself uses the converted Cartesian geometry, not the imported Z-matrix.

The coordinate engine assigned the planar C_{2v} model and generated 36 final non-redundant symmetry-assigned GICs. Thirteen A_1 coordinates were active: eight stretching-type and five bending-type coordinates. The planar-component analysis selected the AB pair, i.e., (I_a, I_b) in the moment formulation; the C constants were recomputed from the optimized Cartesian structure and retained as diagnostics.

The fit used nine isotopologues and reports 27 corrected rotational constants, but only the 18 independent AB moment observables entered the normal equations. It converged in four accepted steps without rejections. The final optimized-space Hessian is rank deficient only in the statistical sense expected for this highly correlated planar data set, while all fitted directions have positive curvature. The RMS rotational residual over all reported constants is 0.1480 MHz, with a maximum of 0.2605 MHz for the diagnostic C component of the parent species. Forced AC and BC fits give larger RMS residuals of 0.2593 and 2.4602 MHz, respectively.

5.7 Azulene Comparison with the MSR Refinement

Azulene is a more demanding polycyclic validation because the MSR analysis uses a step-wise strategy and selected fixed parameters to compensate for incomplete deuterium substitution. The `SEfit` calculation used the same spectroscopic information and structural constraints, but no Z-matrix was built. The MSR-fixed C1–H11/C3–H13 and C2–H12 distances, together with selected fixed valence angles, were supplied as primitive constraints and projected onto the active space.

For the MSR-comparable subset, the diagnostic table contains 12 isotopologues and 36

Table 10: Nitrobenzene planar semiexperimental benchmark. The fit used the independent AB moment pair; rotational residuals are reported over all diagnostic A, B, and C constants in MHz.

Quantity	Value
Isotopologues	9
Reported corrected constants	27
Fitted independent components	AB / (I_a, I_b)
Final non-redundant GICs	36
Active A_1 GICs	13
Numerical rank	13
Accepted/rejected steps	4/0
Condition number	27464.44
Weighted RMS in inertia space	1.32×10^{-2}
RMS rotational-constant residual	0.1480
Maximum rotational-constant residual	0.2605
Reduced χ^2	6.26×10^{-4}
Stationary point	flat or rank deficient

corrected rotational constants. Because azulene is planar, only the AB pair, corresponding to (I_a, I_b), entered the normal equations; the C constants were retained as diagnostics. The coordinate engine generated 48 non-redundant symmetry-assigned GICs. Seventeen were totally symmetric, and the primitive constraints reduced the effective least-squares space to 10 variables. The topology-validating trust-region solver retained the C_{2v} coordinate model throughout the fit, and the final point was a minimum. The RMS residual over all reported constants was 0.0731 MHz, with a maximum of 0.2900 MHz for the A component of azulene-9-^{13}C. Forced AC and BC planar pairs give larger RMS residuals of 0.1086 and 0.3232 MHz.

Cyclopentadiene, azulene, and norcamphor were also used to compare the default GIC coordinate model with the symmetry-Cartesian model. The comparison is summarized in Table 11. In each case, the Cartesian basis spans the same vibrational space as the non-redundant GIC basis after removal of translations and rotations, and the totally symmetric block has the same effective rank after projection of primitive constraints. The rotational residuals are unchanged within numerical precision. The condition number is somewhat system dependent: the GIC model is slightly better conditioned for cyclopentadiene, whereas the symmetry-Cartesian model improves the azulene and norcamphor conditioning. Since the final reported structural parameters are evaluated from the optimized Cartesian geometry and their uncertainties are propagated from the fitted covariance matrix, the lack of direct chemical locality in the working Cartesian amplitudes is not a practical limitation for the final structural output.

Table 11: Comparison of `SEfit` working-coordinate models. "Work." is the number of non-redundant GICs or symmetry-adapted Cartesian vibrational directions, "Sym." is the number of totally symmetric working coordinates before primitive-constraint projection, and "Eff." is the number of independent least-squares variables after projection. Residuals are RMS/maximum absolute values in MHz.

System	Coordinate model	Work.	Sym.	Eff.	Rank	Cond.	Residuals
Cyclopentadiene	GICs	27	10	10	10	454.5	0.0635/0.2289
Cyclopentadiene	Sym. Cartesians	27	10	10	10	635.4	0.0635/0.2289
Azulene	GICs	48	17	10	10	956.7	0.0731/0.2900
Azulene	Sym. Cartesians	48	17	10	10	744.8	0.0731/0.2900
Norcamphor	GICs	48	48	18	18	3700.3	0.0710/0.2091
Norcamphor	Sym. Cartesians	48	48	18	18	1615.8	0.0710/0.2091

Selected structural parameters are compared with the reference MSR values in Table 12. The goal is not to reproduce the MSR Z-matrix parameterization, but to verify that the Cartesian/GIC refinement gives the same structural scale without coordinate-path choices. In the original Z-matrix treatment, one ring C–C distance is a ring-closure quantity derived from the remaining coordinates; here all topological bond lengths are evaluated uniformly from the final Cartesian geometry. For the strongly correlated $R(1,9)/R(4,5)$ pair, the MSR values are mapped to the corresponding Cartesian/GIC bonds before differences are formed. The complete parameter comparison and rotational-constant table are in the Supporting Information.

5.8 Norcamphor Bridged-Topology Stress Test

Norcamphor is a non-planar bridged-molecule stress test combining C_1 topology, a carbonyl group, fused ring constraints, and no deuterium-substituted isotopologues. The input geom-

Table 12: Selected azulene structural parameters from the automatic Cartesian/GIC refinement compared with the reference MSR values. Bond lengths are in Å; angles are in degrees. The reported `SEfit` uncertainty is one standard deviation propagated from the optimized active GIC covariance matrix.

Parameter	`SEfit`	σ	MSR	Difference
$R(1,9)$	1.39745	0.00599	1.39900	-0.00155
$R(4,10)$	1.37923	0.01201	1.37800	+0.00123
$R(4,5)$	1.40122	0.00653	1.40100	+0.00022
$R(5,6)$	1.39054	0.00442	1.39200	-0.00146
$R(4,14)$	1.08356	0.00220	1.08700	-0.00344
$R(5,15)$	1.08133	0.00230	1.08200	-0.00067
$R(6,16)$	1.08117	0.00441	1.08200	-0.00083
$A(1,9,10)$	106.13001	0.52104	106.50000	-0.36999
$A(5,6,7)$	130.09492	0.17695	129.90000	+0.19492
$A(6,5,15)$	115.90333	0.30686	115.70000	+0.20333
$A(4,10,9)$	127.46958	0.08847	127.58000	-0.11042
$A(5,6,16)$	114.95254	0.08847	115.07000	-0.11746

etry was a computed Cartesian structure derived from the published Supporting Information geometry. The isotopologue data consisted of the parent species, all seven singly substituted ^{13}C species, and the ^{18}O isotopologue. The ground-state rotational constants were corrected with computed vibrational contributions using the additive convention $B_e = B_0 + \Delta B_{\text{vib}}$.

Because norcamphor is C_1, symmetry does not reduce the active space: all GICs are totally symmetric. The coordinate engine generated 48 final non-redundant GICs, comprising 18 stretches, 25 bends, and 5 torsional/ring coordinates. The reduced-dimensionality MSR model fixes hydrogen positions through Z-matrix variables; here the same information was translated into 30 primitive constraints, one C–H stretch, one C–C–H angle, and one C–C–C–H dihedral per hydrogen. Projection of these constraints reduced the least-squares space from 48 to 18 variables, corresponding to the heavy-atom skeleton.

The equal-weight moment-of-inertia refinement converged in eight accepted steps without rejections, and the final point was a minimum. The RMS residual over the 27 corrected rotational constants was 0.0710 MHz, with a maximum of 0.2091 MHz. The published reduced-dimensionality MSR refinement reports a standard deviation of 0.05 MHz for the reduced-dimensionality reference model. Here the residuals are dominated by the A constants of the ^{13}C1 and ^{13}C2 isotopologues, while the residual mean for each rotational component remains close to zero. Because the present benchmark uses the printed vibrational corrections, rounded to 0.1 MHz, the remaining difference should not be overinterpreted. A fully homogeneous comparison would require unrounded corrections and the exact statistical convention of the original MSR run. An automatic local-frame hydrogen constraint variant gives a slightly smaller RMS residual of 0.0673 MHz, but the tabulated comparison uses the Supporting Information (SI) Z-matrix-equivalent primitive constraints.

As an independent geometrical diagnostic, Kraitchman substitution coordinates were computed for the singly substituted heavy-atom isotopologues and compared with the absolute coordinates of the fitted structure in the parent principal-axis frame. This comparison was not used in the least-squares objective. It provides a direct check that the fitted Cartesian geometry is consistent with the substitution information. The RMS absolute-coordinate difference over the 24 axis components is 0.02097 Å, and the largest component difference is 0.06231 Å for the a coordinate of the ^{13}C1 isotopologue (Table 13).

Table 13: Kraitchman diagnostic for the norcamphor refinement. RMS and maximum deviations compare absolute Kraitchman substitution coordinates with absolute fitted coordinates over the a, b, and c principal axes. Values are in Å.

Isotopologue	Atom	RMS	Max.	Axis
^{13}C1	C1	0.03716	0.06231	a
^{13}C2	C2	0.02043	0.02777	b
^{13}C3	C3	0.01640	0.02080	c
^{13}C4	C4	0.02469	0.04217	c
^{13}C5	C5	0.02139	0.03468	b
^{13}C6	C6	0.01452	0.02268	c
^{13}C7	C7	0.00638	0.00769	a
^{18}O	O8	0.01154	0.01591	c

The symmetry-Cartesian refinement was repeated with the same 30 primitive hydrogen constraints. Because norcamphor is C_1, all 48 vibrational directions are totally symmetric before constraint projection; the constraints again reduce the fit to 18 effective variables. The Cartesian calculation converges in eight accepted steps to a minimum, gives the same rank and RMS/maximum rotational residuals of 0.0710/0.2091 MHz, and lowers the condition number from 3700.3 to 1615.8.

Table 14: Norcamphor C_1 reduced-dimensionality stress test. Moments of inertia were used as fit observables; rotational residuals are reported in MHz.

Quantity	Value
Isotopologues	9
Rotational observables	27
Final non-redundant GICs	48
Totally symmetric GICs	48
Primitive hydrogen constraints	30
Effective independent variables	18
Numerical rank	18
Accepted/rejected steps	8/0
Condition number	3700.28
Weighted RMS in inertia space	4.94×10^{-3}
RMS rotational-constant residual	0.0710
Maximum rotational-constant residual	0.2091
Reduced χ^2	7.31×10^{-5}
Stationary point	minimum

5.9 Refinements with Parameter Classes

Parameter classes provide a practical strategy for semiexperimental refinements in which the available isotopic information is insufficient to determine all structural variables independently. Hydrogen-containing distances or angles can be refined collectively or held fixed as symmetry-complete primitive-coordinate relations. The black-box character of the workflow therefore does not eliminate chemical judgment; it confines that judgment to constraints, predicates, and classes that have an unambiguous internal-coordinate meaning.

6 Conclusions

The central result is a clean separation of roles. Molecular structures are supplied and reported in Cartesian and primitive internal coordinates. Constraints, predicates, and parameter classes are defined in the same primitive layer. The least-squares solver acts only on a non-redundant totally symmetric active space generated from topology and symmetry.

Two active coordinate choices were tested. Symmetry-adapted GICs provide a chemically local internal-coordinate basis because redundancy removal and symmetry adaptation are performed within homogeneous families. Symmetry-adapted Cartesians provide a direct Z-matrix-free basis in which only translations, rotations, and non-totally-symmetric displacements are removed. Although Cartesian amplitudes are not local chemical parameters, this is not a limitation because final bond lengths, angles, dihedrals, and uncertainties are evaluated from the optimized Cartesian structure.

The benchmarks show where this distinction matters. Glycolaldehyde is a standard acyclic molecule for which a conventional Z-matrix is simple. Glycine II requires functional constraints to compensate for incomplete isotopic information, whereas cyclopentadiene, nitrobenzene, azulene, and norcamphor require less uniform manual choices in Z-matrix workflows because of ring closure, planar rotational redundancy, fused or bridged topology, symmetry, and missing deuterium information. Whenever both coordinate models are used, the GIC and symmetry-Cartesian refinements give the same effective dimensions, ranks, and rotational residuals. The planar nitrobenzene and azulene tests show that only two rotational components should enter the normal equations; the omitted component is a diagnostic consistency check. In norcamphor, the Cartesian model also lowers the condition number from 3700.3 to 1615.8 while preserving the 18-dimensional constrained heavy-skeleton model.

The broader conclusion is that robust refinement of experimental data to molecular structures does not require a user-designed Z-matrix. It requires a complete non-redundant symmetry-consistent active space, chemically meaningful constraints expressed in internal coordinates, and rigorous covariance propagation to the structural quantities of interest. `SEfit` implements this strategy as a reproducible workflow that can be reused by other coordinate-dependent modules.

In this form, `SEfit` is positioned as the natural successor to MSR. The statistical core of MSR, automated starting-geometry generation, fast vibrational-correction workflows, and reduced-dimensionality treatments of missing C–H deuteration are retained. The coordinate model, however, is no longer a manual Z-matrix: it is generated from topology, constrained through primitive internal coordinates, exposed through a more flexible interface, and reported with uniform propagated uncertainties. The same code base separates covalent topology from auxiliary reference information, so model-dependent geometry corrections do not alter rank, ring perception, or GIC symmetry classification.

Supporting Information Available

Complete glycine II, cyclopentadiene, nitrobenzene, and azulene rotational-constant comparisons, including corrected experimental constants, calculated constants, and differences. The nitrobenzene Kraitchman diagnostic is also provided. Additional diagnostic tables for glycolaldehyde and norcamphor, including final Cartesian coordinates, propagated topological parameters, and residuals, are generated by the archived `SEfit` output.

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